

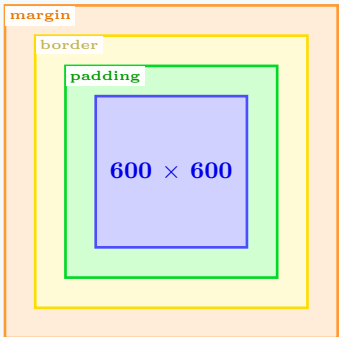
Architectural Severance via object-fit: cover

Decoupling the HTML Box Model from the Raster Graphic Payload



CSS Box Model

HTML Tag



Viewport-Driven Dimensions

Defined by responsive CSS:

width: 100%;

aspect-ratio: 1 / 1;

(Layout engine creates an empty void)



Intrinsic Payload

Raster Image File



Pixel-Driven Dimensions

Defined by file metadata:

Intrinsic Width: 2000px

Intrinsic Height: 1000px

(Unaffected by DOM layout)

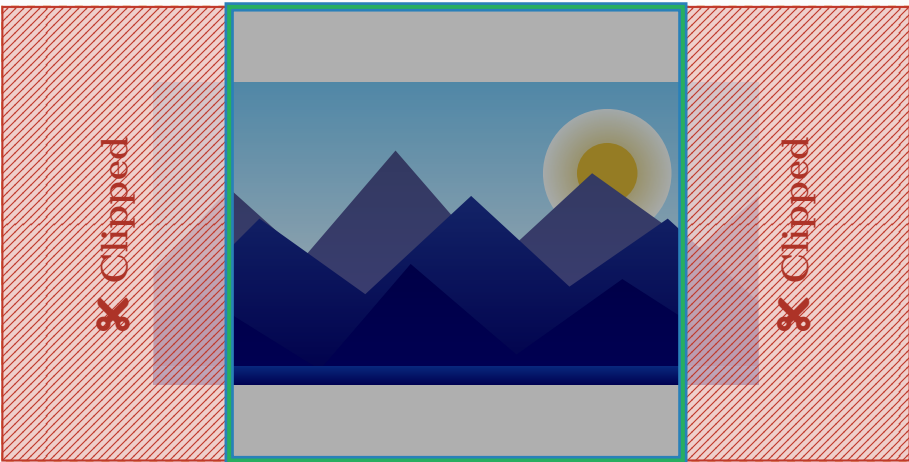


Render Engine Compositing (The “Severance”)

2. Clip-Path Isolation

The DOM container acts purely as a *clipping mask* or window. Overflowing pixels (red zones) are discarded from the render tree, entirely severing layout dependency.

Content Box Boundary (600px)



✓ Final Painted Output (A perfect square)

1. Proportional Scale Matrix

The engine computes a transform matrix to scale the intrinsic payload so both W and H completely cover the container void.

$$\max \left(\frac{W_{dom}}{W_{img}}, \frac{H_{dom}}{H_{img}} \right)$$